Quantized reward-model verifiers in Best-of-N mathematical answer selection: accuracy, memory and value retention

Hector Borobia[a,*], Elies Seguí-Mas[b], Guillermina Tormo-Carbó[c]

[a] *VRAIN – Valencian Research Institute for Artificial Intelligence, Universitat Politècnica de València, Valencia, Spain*
[b] *Department of Economics and Social Sciences, Universitat Politècnica de València, Valencia, Spain*
[c] *Department of Business Organisation, Universitat Politècnica de València, Valencia, Spain*

Abstract

Reward-model verifiers improve Best-of-N answer selection but compete with the generator for accelerator memory. We evaluate post-training weight quantization of verifiers on frozen pools of 32 solutions per problem from GSM8K and MATH, using three outcome reward models (1.7B–8B) from one family and a 7B process reward model (PRM), with GPTQ, AWQ, NF4 and round-to-nearest. Analysis plans were fixed in version control before verifier scoring, and for two further studies before candidate generation. On its prespecified 1,200-problem population, the 8B GPTQ-W4 contrast meets a ±3-point equivalence bound for Best-of-32 accuracy (difference +0.33; 90% interval −0.58 to +1.25). Both further studies, on unused MATH problems, support GPTQ-W4 equivalence for the 8B outcome verifier and the PRM.

*Corresponding author.
Email addresses: hecboar@doctor.upv.es (Hector Borobia), eseguim@upvnet.upv.es (Elies Seguí-Mas), gtormo@omp.upv.es (Guillermina Tormo-Carbó)

Where both verifiers beat majority vote, the PRM retains an estimated 93% of its 8.2-point advantage and passes the prespecified half-retention test. With a stronger generator, neither verifier clearly beat majority vote. Two Llama-based reward models also stay within the equivalence bound (losses 0.6–1.5 points). Under a verifier-only 6 GiB budget, a packed 4-bit 4B outcome verifier outperforms the tested 1.7B full-precision verifier by 2.25 points under strict grading and 1.40 points after a human audit. Four-bit quantization often changes the selected candidate but much less often its answer. Three-bit GPTQ with generic calibration degrades selection; in-domain calibration recovers most of the loss where it was large. Strict answer grading can turn reward-model format preferences into spurious quantization gains, so compressed verifiers need format-robust grading and a majority-vote baseline.
Keywords: post-training quantization, reward models, process reward models, Best-of-N selection, GPTQ, majority voting, equivalence testing

1. Introduction

Sampling several candidate solutions and letting a learned verifier pick one (Best-of-N) is one of the simplest ways to turn extra inference compute into accuracy on mathematical reasoning [1, 2, 3]. The verifier is usually a reward model with a few billion parameters. An outcome reward model (ORM) scores a complete response, and a process reward model (PRM) scores each reasoning step. In deployment the verifier competes with the generator for accelerator memory, and an obvious way to make room is post-training quantization (PTQ) of its weights to 4 bits or fewer [4, 5, 6].

It is not obvious that this preserves selection quality. PTQ is usually

evaluated by perplexity or by the accuracy of the quantized model on its own task. A verifier, by contrast, only has to *rank* candidates so that a correct one ends up on top. Small score perturbations do no harm as long as the winning answer stays the same, but Best-of-N is sensitive to errors at the top of the ranking [7]. One might therefore expect quantization damage to grow with the budget N and with problem difficulty, and to matter most for small verifiers. If, on the other hand, quantized verifiers select as well as full-precision ones, a fixed memory budget may be better spent on a larger quantized verifier.

We study these questions with frozen candidate pools and analysis plans specified in advance. Each pool of 32 candidates per problem is generated once, so every verifier and precision ranks exactly the same responses, and any difference in accuracy comes from the verifier. We evaluate three ORM sizes and one PRM in BF16, NF4, GPTQ (4 and 3 bits), AWQ and round-to-nearest on 2,000 GSM8K and MATH problems, and we measure verifier memory with packed 4-bit kernels. Two further studies, H6 with a stronger generator and H7 with new hard problems, had their plans fixed before their candidates were generated.

Our contributions are as follows.

1. **Accuracy equivalence at 4 bits.** GPTQ-W4 meets a prespecified ± 3 pp equivalence bound for the Best-of-32 accuracy of an 8B ORM (90% interval $[-0.58, +1.25]$), and the result holds for both an ORM and a PRM in H6 and H7. Three-bit GPTQ with generic calibration degrades selection, most severely for the smallest Qwen-based verifier and for two Llama-based reward models. In exploratory tests, calibrating

on in-domain candidates reduced the loss from 13.2 to 3.8 pp for the 1.7B ORM and from 12.6 to 2.8 pp for a Llama-based ORM, while a length-matched generic control did not help.

2. **Retained verifier value.** In H7, on 1,060 new hard problems where both verifiers beat majority vote, a prespecified test shows that 4-bit quantization keeps more than half of their advantage. The PRM retains an estimated 93% of an 8.2 pp advantage and the ORM 91% of a 3.0 pp advantage, the latter narrowly significant. With a strong generator, neither verifier clearly beat majority vote.
3. **A tested memory comparison.** Under a 6 GiB verifier-only budget and a fixed workload, a 4B ORM with packed 4-bit weights outperforms the best tested full-precision verifier that fits. The difference is 2.25 pp under the prespecified grader and 1.40 pp after a blinded human audit of the winners without an extractable answer.
4. **Candidate-level instability without accuracy loss.** At 4 bits, quantization changes the selected candidate on a quarter to a half of problems but changes the selected answer far less often. Harm and benefit stay small and largely cancel, which is consistent with answer redundancy in the pools.
5. **Evaluation lessons.** A strict answer grader can turn reward-model format preferences into spurious quantization gains. We show how a format/content decomposition, a lenient grader, a blinded human audit and revised labels detect and correct this. We also report two prespecified hypotheses that the evaluation data did not support: that damage concentrates on MATH and that it grows with N.

2. Related work

Verifiers and test-time compute. Cobbe et al. [1] trained a verifier to rank sampled solutions to grade-school mathematics problems. Process supervision, which scores every reasoning step, outperforms outcome supervision on MATH [2], and it was followed by automatically labelled PRMs [8, 9, 10] and generative verifiers [11]. Several studies ask how inference compute should be split between sampling and verification [3, 12, 13]. They evaluate verifiers at full precision and treat the verifier's own cost as fixed.

Reward models against self-consistency. Majority vote over sampled answers (self-consistency) needs no verifier [14], and its coverage and accuracy keep growing with the number of samples [15, 16]. Whether a reward model beats it depends on the generator, the reward model and the aggregation rule [17, 18], and calibrated combinations of answer frequency and PRM scores can outperform both [19]. Kim et al. [20] study memory-optimal test-time strategies for reasoning models. They include a Best-of-N experiment with an external 7B PRM counted at full precision (13.28 GB; Section 4.1 of the ICLR 2026 version) and find it less memory-efficient than majority voting. Their main analysis quantizes the generator and leaves verifiers out by design. We ask the complementary question of how much of a verifier's cost can be removed by quantizing the verifier itself, and whether its selections survive.

Cheaper verification. Other work reduces the cost of Best-of-N verification without compressing the reward model. It rejects low-scoring generations early [21], learns lightweight rewards from the generator's hidden states [22], or scores from the generator's key–value cache instead of re-encoding the text

[23]. These methods change what is computed. Weight quantization keeps the verifier and its inputs as they are and only lowers the precision of the weights, so the two can be combined.

Reward-model reliability. Reward-model quality is usually measured on preference and step-error benchmarks [24, 25, 26], and Best-of-N is known to over-optimise imperfect reward models [7]. We do not ask whether a reward model is accurate. We ask whether a compressed copy of it selects as well as the original.

Post-training quantization of language models. Post-training quantization reduces memory with little loss of perplexity or task accuracy at 8 bits [27, 28] and, for weights only, at 4 bits [4, 5, 6]. For a fixed bit budget 4-bit precision is close to optimal [29], and large evaluations find that well-calibrated 4- and 8-bit models track their full-precision versions [30]. The topic is also active in this journal [31, 32]. The choice of calibration data matters as well: Doubt-Preserving Quantization (DPQ) builds calibration sets that mix examples on which the full-precision model is uncertain with generic anchor text, in order to preserve its confidence and decision margins [33]. For reasoning, quantizing the *generator* hurts more at 3 bits and on harder problems [34, 35]. Aggregate accuracy can also hide many changed outputs, since compressed models can keep their accuracy while flipping a sizeable share of individual answers [36]. We measure the corresponding effect for selection by counting how often quantization changes the selected candidate and how often it changes the selected answer.

Quantized evaluators. The closest work quantizes LLM-as-judge models. A concurrent workshop paper [37] reports that INT8 judges keep their agreement with human labels within a preregistered equivalence region on pairwise chat comparisons, while NF4 flips a quarter to a third of verdicts and lowers one judge's agreement. We study scalar reward models used for Best-of-N selection on problems with verifiable answers. We also compare several quantization methods, measure memory with packed kernels, and test the verifier's advantage over majority vote. We did not find prior work that quantizes the verifier in Best-of-N and measures the effect on selection.

Evaluation practice. Math benchmark results are sensitive to answer extraction, contamination and label errors [38, 39, 40, 41], and confidence intervals are still uncommon in model evaluations [42]. Preregistration has been proposed for predictive modelling [43] but remains rare in machine learning. We fix the analysis plans in advance in version control and report uncertainty for every contrast. We also check the outcome measure with a lenient grader, a human audit and revised labels.

3. Experimental setup

3.1. Analysis plans

The design, estimands and tests were specified in advance in a version-controlled repository, and Table C.8 in Appendix C gives the chronology. For the main study, the initial plan and its first amendment preceded the generation of the evaluation candidates. The final amendment came after we had inspected generator-side statistics (accuracy, share of extractable answers, truncation) and before any verifier scored an evaluation problem. The plans

of the two further studies, H6 and H7, with their problem manifests and analysis code, were fixed before their candidates were generated. Because we did not use an external registry, the commit history is the timestamped record of each plan. The hypotheses are labelled c1–c5 in the main study, H6a/b in H6 and H7a/b in H7. Every result outside these tests is exploratory and labelled as such.

3.2. Problems and candidate pools

Table 1 lists the five evaluation populations, and each row is used as follows. *Main, full* is the main evaluation set of 2,000 problems: 800 from the GSM8K test split [1], 400 sampled from MATH500 [2], and 800 further MATH test problems [44] outside MATH500, sampled with a fixed seed. It is used for the memory comparison c2 and for the exploratory grid. *Main, original manifest* is its first 1,200 problems (GSM8K and MATH500). This subset was frozen before the 800 extra MATH problems were added, and it is the prespecified population of c1. *Main, MATH* contains the 1,200 MATH problems of the main set (MATH500 and the extension) and is used for c4 and c5. *H6* (800 MATH Level-5 test problems) and *H7* (1,060 MATH Level-4 and Level-5 test problems) are the populations of the two further studies; none of their problems appears anywhere else in this work. A separate development set of 400 problems (200 GSM8K train, 200 MATH train) was used only for the go/no-go gate and for the choices assigned to development data.

The main generator is Qwen2.5-1.5B-Instruct [45]. For every problem it produces a pool of 32 candidates (4 seeds $\times$ 8 samples, temperature 0.8, top-p 1.0, at most 1,024 new tokens) with the system prompt "Please reason step by step, and put your final answer within `\boxed{}`." The pools are

Table 1: Principal evaluation populations. "New" problems appear nowhere else in this work (text-level exclusion against all other items and against all of MATH500).

population	n	composition	used for
Main, full	2,000	800 GSM8K + 400 MATH500 + 800 MATH test	c2, Table 3
Main, original manifest	1,200	800 GSM8K + 400 MATH500	c1
Main, MATH	1,200	400 MATH500 + 800 MATH test	c4, c5
H6	800	MATH test Level 5 (new)	H6a/b
H7	1,060	MATH test Level 4 (905) + Level 5 (155), new	H7a/b

generated once in BF16 with vLLM 0.30 [46], then frozen and hashed, so every verifier and precision scores exactly the same candidates. Candidate accuracy is 52.1%, 88.7% of candidates have an extractable final answer, and 6.5% reach the length limit. A secondary generator, Llama-3.2-3B-Instruct, produces pools for a 300-problem subset (exploratory; Section 4.7).

3.3. Verifiers

The main study uses three outcome reward models (ORMs) from one family, Skywork-Reward-V2-Qwen3 at 1.7B, 4B and 8B parameters [47], and one process reward model (PRM), Qwen2.5-Math-PRM-7B [10]. We chose the Skywork family because it provides several sizes on one backbone, all trained on the same curated preference data [47], which lets us compare size and precision within a family, and because they reported state-of-the-art results on seven reward-model benchmarks at release. The Skywork models are general preference reward models, trained on broad human-preference data rather than on mathematical correctness. We use them as outcome verifiers by scoring the full (problem, response) pair with their scalar head. The PRM, by contrast, is trained for mathematical step correctness, so the

study covers both kinds of supervision. Its step rewards are aggregated by their minimum, and the last-step reward is reported as a sensitivity analysis. To test generality beyond the Skywork-Qwen family, two further ORMs on a different backbone are evaluated in exploratory experiments: Skywork-Reward-V2-Llama-3.1-8B (same training-data family, Llama 3.1 backbone) and Llama-3.1-Tulu-3-8B-RM [48] (different laboratory and data). All models are run with their documented input formats (Appendix A).

3.4. Precisions and deployments

Quantization is weight-only. The weights of all linear layers inside the transformer blocks are quantized, while embeddings, norms, the reward head, activations and computation stay in BF16. "BF16" denotes the uncompressed model. The conditions are:

- **NF4**: 4-bit NormalFloat [6], both simulated and deployed with `bitsandbytes` (bit-identical scores);
- **GPTQ-W4** and **GPTQ-W3**: GPTQ [4] with group size 128, calibrated on 128 WikiText-2 sequences of 512 tokens (generic calibration);
- **AWQ-W4** [5] with the same calibration, and round-to-nearest **RTN-W4** and **RTN-W8** for an exploratory dose-response.

Accuracy experiments use simulated ("fake") quantization, in which weights are quantized and then dequantized back to BF16. The scores then differ from BF16 only by the rounding of the weights, so every method and bit width runs on the same kernels and the comparison is not mixed with kernel-specific numerics. Simulation also covers conditions that have no packed deployment

here (GPTQ-W3, AWQ, RTN and the PRM). We check that it represents real deployment by comparing it with packed GPTQ-W4 kernels (Section 4.7). For the memory study, the GPTQ-W4 ORMs are exported and run with packed 4-bit kernels (`gptqmodel`, Marlin). The PRM has no packed GPTQ deployment because the export does not support its remote modelling code, so its deployed 4-bit condition is NF4. Every quantized model is audited: every target layer must differ from BF16 and every other parameter must stay bit-identical.

Verifier memory is the peak allocated GPU memory, reported in GiB (2^{30} bytes; 1 GiB $\approx$ 1.074 GB, where 1 GB = 10^9 bytes). GB appears only where a source states it, such as GPU specifications. Memory is measured while scoring a fixed workload of 64 sequences of 900 tokens on an RTX 4090 with no generator resident.

3.5. Selection rules and baselines

Best-of-N selects the highest-scoring candidate among N, and exact score ties are broken by a fixed random priority (seed 11) shared by all conditions. For $N < 32$, accuracy is averaged over 200 random subsets per problem, drawn identically for every condition (seed 7); $N = 32$ is the full pool. We report two verifier-free baselines. Majority vote is a plurality vote over canonical answer clusters, formed by numeric normalisation followed by symbolic equivalence with `math_verify`. Candidates without an extractable answer do not vote, and a pool with no extractable answer counts as incorrect. The most frequent cluster wins, with ties broken in favour of the cluster that appears first in candidate order, and the vote is correct if the winning cluster contains a correct candidate. Majority vote therefore does not depend on the verifier.

The oracle (pass@32) counts a problem as solved if at least one of its 32 candidates is correct. It is the upper bound for any rule that selects one candidate from the pool.

3.6. Grading

The prespecified grader extracts the last `\boxed{}` answer, or failing that a GSM8K-style "####" line or an explicit "the answer is" phrase. It normalises numbers and compares MATH answers symbolically. A strict grader can confound verifier preferences with answer format, so we add three checks. The first is a lenient grader, used for sensitivity analyses only, which takes the last number or expression in the final lines when no answer is extractable. The second is a blinded human audit (Appendix B), and the third re-grades GSM8K with the revised GSM8K-Platinum labels [39]. The endpoint throughout is final-answer correctness.

Under the strict grader a winner without an extractable answer always counts as wrong, so Best-of-N accuracy factorises as $A = p\,a$, where p is the share of winners with an extractable answer and a the accuracy among them. A change from BF16 (subscript 0) to a quantized condition (subscript 1) then splits exactly into a format term and a content term, $\Delta = (p_1 - p_0)\,a_0 + p_1\,(a_1 - a_0)$.

3.7. Metrics and controls

For verifier v and condition c, A_N is Best-of-N accuracy and $\Delta_N = A_N^c - A_N^{\mathrm{BF16}}$, computed per problem and averaged. At $N = 32$ we also compare each quantized selection with the BF16 selection of the same problem and report four shares of problems: the selected candidate changes; the selected

canonical answer changes; a correct BF16 selection becomes wrong (harm); and a wrong BF16 selection becomes correct (benefit). Δ_{32} equals benefit minus harm. Two controls measure how much the selection moves without any quantization. *bf16-null* rescores in BF16 with a different batch layout, which changes padding and the order of floating-point operations and hence perturbs scores slightly. *bf16-repeat* rescores with the same layout. The repeat is bit-identical. The null control changes the selected candidate on 7–11% of main-study problems with $|\Delta_{32}| \leq 0.3$ pp, and on 10.9% and 15.0% (H6) and 5.5% and 8.7% (H7) of problems for the ORM and the PRM.

3.8. Statistics

All inference uses a paired percentile bootstrap over problems with 10,000 resamples (seed 7). Problems are resampled with replacement from the full population. Every quantity computed on a problem (both conditions, all verifiers entering a contrast, the majority-vote outcome) is resampled jointly, which keeps within-problem correlations.

- *Differences* are reported with 95% percentile intervals and two-sided centred-bootstrap p-values.

- *Equivalence* at the prespecified ± 3 pp margin is tested by two one-sided tests (TOST), $p_{\text{TOST}} = \max\{P^*(\bar{d} \leq -0.03),\ P^*(\bar{d} \geq 0.03)\}$. The test inverts the percentile interval, so $p_{\text{TOST}} \leq \alpha$ if and only if the $(1 - 2\alpha)$ interval lies inside the margin. We checked its size by simulation with the true mean on either margin, drawing $n = 800$ per-problem differences from $\{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$ with probabilities $(0.075, 0.03, 0.82, 0.03, 0.045)$ for the lower margin and the mirror image for the upper one (2,000

replicates of 2,000 resamples). The rejection rates were 0.056 and 0.054 at nominal 0.05, 0.024 and 0.025 at 0.025, and 0.006 and 0.004 at 0.0125, with a Monte Carlo standard error of about 0.005. The ± 3 pp margin was fixed in the plan before any evaluation score existed. On the c1 population it corresponds to a 13% relative increase in errors.

- *Value retention* (H6b, H7b) is tested one-sided, $p = P^*(\bar{R} \leq 0)$ (Section 4.5).

- *Resolution.* Bootstrap p-values are Monte Carlo estimates with resolution 10^{-4}. When no resample crosses the boundary we report the count $(0/10^4)$ instead of $p = 0$, and the corresponding 95% Monte Carlo upper bound is $3 \cdot 10^{-4}$.

- *Multiplicity* is controlled by Holm correction within each prespecified family (c1–c5, H6a/b and H7a/b). The k-th smallest of m raw p-values is multiplied by $m - k + 1$ and the cumulative maximum is taken. A test of the family that was not run keeps $p = 1$. In the main-study family, c1 enters with the two-sided p-value of its difference test, as implemented when the family was fixed. Its decision rule is equivalence, judged by whether its 90% interval lies inside ± 3 pp (Table 2).

The five main-study contrasts are the following. c1 is the equivalence of Δ_{32} for the 8B ORM at GPTQ-W4 on the original 1,200-problem manifest. c2 compares the deployment selected on development data with the best BF16 deployment under the 6 GiB verifier memory ceiling. c3 compares CH-GPTQ, a GPTQ calibration with a contrastive Hessian term built from winner–challenger pairs, with an adapted DPQ recipe [33]; it was not run

(Appendix C). c4 is Δ_{32} on the 1,200 MATH problems for the 4B and 8B ORMs at GPTQ-W4, averaged within problem, and c5 is the change of that loss from $N = 4$ to $N = 32$.

4. Results

Table 2 summarises every confirmatory test. The subsections below give the details and the exploratory analyses around them.

4.1. Four-bit quantization preserves Best-of-32 accuracy

The prespecified contrast c1 uses the 8B ORM at GPTQ-W4 on the original 1,200-problem manifest, where BF16 accuracy is 77.5%. Quantization changes accuracy by $\Delta_{32} = +0.33$ pp, with a 90% interval of $[-0.58, +1.25]$. The interval lies inside the prespecified ± 3 pp margin, and no bootstrap resample out of 10,000 fell beyond either margin, so **equivalence is supported**. On this population a 3 pp loss would have meant 13% more errors. On the full 2,000 problems the same contrast is $+0.15$ pp $[-0.65, +0.95]$ (95% interval; Table 3).

Table 3 gives the exploratory grid on the 2,000 problems. At 4 bits, GPTQ-W4 and NF4 change A_{32} by between -1.1 and $+0.6$ pp for the three ORMs and the PRM, even though the selected candidate changes on 21–49% of problems. The 4-bit curves track BF16 at every budget $N \leq 32$ (Fig. 1). Selection degrades at 3 bits with this recipe. GPTQ-W3 costs 1.3–1.8 pp for the 4B ORM, the 8B ORM and the PRM, and 13.2 pp for the 1.7B ORM, whose accuracy falls towards the random-candidate rate. For the 4B and 8B ORMs the quantization method makes little difference at 4 bits. Their Δ_{32} is 0.00 and $+0.55$ pp with AWQ-W4, $+0.25$ and $+0.90$ pp with round-to-nearest

Table 2: All confirmatory tests. (a) Inference: estimates in percentage points with 95% intervals (c1: 90%, its prespecified decision rule). p is the raw bootstrap p-value (two-sided for differences, TOST for equivalence, one-sided for $R > 0$); "$0/10^4$": no resample out of 10,000 crossed the boundary (95% upper bound $3 \cdot 10^{-4}$). p_{H}: Holm within the family. (b) Best-of-32 accuracies (%) under the prespecified grader; MV is the verifier-independent majority vote.

(a) Inference

test	population (n)	contrast	estimate [CI]	p	p_{H}	result
c1	main (1,200)	8B ORM, W4 vs BF16	+0.33 [−0.58, +1.25]	$0/10^4$[a]	–[a]	equivalent
c2	main (2,000)	4B W4 packed vs 1.7B BF16	+2.25 [+1.05, +3.45]	0.0001	0.0005	supported
c3	–	CH-GPTQ vs DPQ-adapted (not run)	–	1	1	not tested
c4	main MATH (1,200)	4B+8B ORM, W4 vs BF16	−0.63 [−1.38, +0.08]	0.099	0.40[b]	not supported
c5	main MATH (1,200)	loss growth, N=4→32	−0.51 [−1.20, +0.16]	0.140	0.42[b]	not supported
H6a	H6 (800)	8B ORM, W4, ±3	0.00 [−1.13, +1.13]	$0/10^4$	$0/10^4$	equivalent
H6a	H6 (800)	PRM, W4, ±3	+0.50 [−0.50, +1.50]	$0/10^4$	$0/10^4$	equivalent
H6b	H6 (800)	8B ORM, $R > 0$	−1.88 [−3.19, −0.63]	0.998	0.998	not supported
H6b	H6 (800)	PRM, $R > 0$	+0.56 [−0.63, +1.69]	0.182	0.365	not supported
H7a	H7 (1,060)	8B ORM, W4, ±3	−0.28 [−1.42, +0.85]	0.0001	0.0002	equivalent
H7a	H7 (1,060)	PRM, W4, ±3	−0.57 [−1.32, +0.19]	$0/10^4$	$0/10^4$	equivalent
H7b	H7 (1,060)	8B ORM, $R > 0$	+1.23 [−0.14, +2.59]	0.041	0.041	supported
H7b	H7 (1,060)	PRM, $R > 0$	+3.54 [+2.45, +4.72]	$0/10^4$	$0/10^4$	supported

(b) Accuracies

test	verifier	BF16	quantized	MV
c1	8B ORM	77.5	77.8	78.3
c2	1.7B BF16 / 4B W4 packed	71.8	74.1	72.1
c4	4B and 8B ORM (mean)	67.1	66.5	64.8
H6	8B ORM / PRM	72.8 / 76.6	72.8 / 77.1	76.5
H7	8B ORM / PRM	59.0 / 64.2	58.7 / 63.6	55.9

[a] c1 is judged by equivalence: its 90% interval lies inside ±3 pp, and the TOST p-value is shown. The prespecified Holm family entered c1 with the two-sided p-value of its difference test (0.588, Holm 1.00), and the Holm values of c4 and c5 shown here are computed with it. [b] With the TOST p-value of c1 in the family instead, the Holm-adjusted values of c4 and c5 would both be 0.30; no decision changes.

(RTN-W4), and +0.25 pp with RTN-W8 on the 8B. The 1.7B ORM is more sensitive to the method and loses 2.8 pp $[-4.0, -1.7]$ with RTN-W4, against 1.1 pp with GPTQ-W4.

Table 3: Best-of-32 accuracy (%) and change versus BF16 (pp, 95% CI) in the main study (2,000 problems, Qwen2.5-1.5B-Instruct candidates). Winner change is the share of problems whose selected candidate differs from the BF16 one. Verifier-free baselines: majority vote 72.1%, oracle pass@32 86.9%, random candidate 52.1%.

verifier	precision	A_{32}	Δ_{32} [95% CI]	winner change
Skywork-1.7B	BF16	71.8	–	–
	NF4	70.7	−1.10 [−2.00, −0.25]	34%
	GPTQ-W4	70.7	−1.10 [−2.25, +0.05]	49%
	GPTQ-W3	58.7	−13.15 [−15.05, −11.25]	88%
Skywork-4B	BF16	73.5	–	–
	NF4	72.7	−0.80 [−1.55, −0.10]	24%
	GPTQ-W4	74.1	+0.55 [−0.30, +1.40]	29%
	GPTQ-W3	71.8	−1.75 [−3.05, −0.45]	60%
Skywork-8B	BF16	72.9	–	–
	NF4	72.9	−0.05 [−0.75, +0.65]	21%
	GPTQ-W4	73.1	+0.15 [−0.65, +0.95]	25%
	GPTQ-W3	71.3	−1.60 [−2.75, −0.45]	50%
PRM-7B	BF16	76.6	–	–
	NF4	76.5	−0.05 [−0.65, +0.50]	25%
	GPTQ-W4	76.3	−0.30 [−1.05, +0.45]	33%
	GPTQ-W3	75.3	−1.30 [−2.30, −0.25]	54%

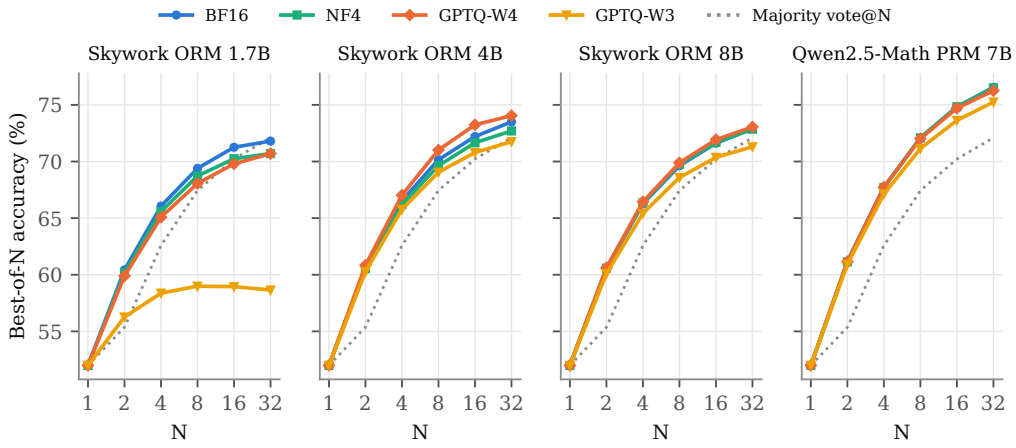


Figure 1: Best-of-N accuracy as a function of the number of candidates N, per verifier and precision (main study, 2,000 problems; 200 random subsets per problem for $N < 32$). The dotted line is the verifier-free majority vote over the same N candidates.

4.2. Under the 6 GiB verifier budget, the tested 4B 4-bit verifier is best

Fig. 2 shows accuracy against measured verifier memory for the tested deployments. The plan states the ceiling as 6 GB, and we apply it to the GiB measurements as 6 GiB. The selected configuration needs 4.92 GiB (5.3 GB) and fits under either reading. The configuration selected on development data is the 4B ORM with packed GPTQ-W4, and the best BF16 verifier that fits is the 1.7B ORM (4.02 GiB). The prespecified contrast c2 is +2.25 pp [+1.05, +3.45] ($p = 0.0001$, Holm $p = 0.0005$), so the hypothesis is **supported**. The 1.7B BF16 verifier is 0.3 pp below majority vote, which needs no verifier, and the 4B GPTQ-W4 verifier is 2.0 pp above it. The comparison applies to the tested checkpoints, the fixed scoring workload and verifier memory only. In this family accuracy does not grow with size (the 4B BF16 model beats the 8B), so the result describes these deployments and does not imply that

larger verifiers are generally better.

Part of the advantage comes from answer format, and its size depends on the selection policy. Three secondary analyses, each applied identically to both arms, give a consistent but smaller effect. With the lenient grader, c2 is $+1.55$ pp $[+0.45, +2.65]$. Restricting Best-of-32 to candidates with an extractable answer gives 75.4% versus 74.0%, a difference of $+1.40$ pp $[+0.35, +2.45]$, and a reward-weighted vote gives 75.5% versus 74.0%, $+1.50$ pp $[+0.45, +2.55]$. Finally, we audited the outcome measure by hand. The first author graded every distinct Best-of-32 winner without an extractable answer in either arm (94 responses), together with 30 random winners with one, without knowing the arm, the verifier or the grader's verdict (Appendix B). Replacing the strict grader by these labels gives $+1.40$ pp $[+0.30, +2.50]$ (75.5% versus 74.1%, nominal $p = 0.011$). This interval treats the single rater's labels as error-free. None of the three analyses belongs to the prespecified family. Together they suggest that the deployment advantage under format-robust grading or selection is about 1.4–1.5 pp, and that the strict-grader estimate is an upper value. The strict-grader contrast also stays positive with revised GSM8K labels ($+2.37$ pp) and after excluding truncated candidates ($+2.20$ pp).

The re-measurement (Table 4) adds the PRM, which needs 16.72 GiB in BF16 and 7.98 GiB with deployed NF4; the deployed NF4 scores are identical to the simulated ones. With the PRM included, the best tested deployment under an 8 GiB budget is the NF4 PRM (76.5%), ahead of every ORM deployment, while the 6 GiB comparison of c2 is unchanged. Scoring throughput with 4-bit weights was 0.8–3.4% lower than with BF16. The

Table 4: Peak verifier memory and scoring throughput on an RTX 4090 (exploratory re-measurement with a fixed workload of 64 sequences of 900 tokens and no generator resident). Peak allocated memory is recorded over one pass. Throughput is averaged over three timed forward passes after one warm-up pass, with GPU synchronisation. The re-measured peaks match the prespecified measurement used for c2 to within 0.001 GiB. The PRM has no packed GPTQ deployment (Section 3.4).

	BF16		NF4 (`bitsandbytes`)		GPTQ-W4 (packed)	
verifier	GiB	seq/s	GiB	seq/s	GiB	seq/s
Skywork-1.7B	4.02	38.0	2.13	37.6	2.07	36.7
Skywork-4B	9.95	16.5	5.08	16.3	4.92	16.0
Skywork-8B	16.97	9.8	7.67	9.7	7.37	9.7
PRM-7B	16.72	11.4	7.98	11.2	–	–

workload is a large prefill batch of 57,600 tokens per pass. With batches this large the forward pass is likely limited by computation rather than by reading the weights, so smaller weights do not speed it up and dequantization adds a small cost; we did not profile this. We did not measure small batches, end-to-end latency or a generator resident on the same GPU. For scale, the generators' BF16 weights alone take about 2.9 GiB (Qwen2.5-1.5B), 6.0 GiB (Llama-3.2-3B) and 14.2 GiB (Qwen2.5-Math-7B), computed from their parameter counts and excluding the key–value cache.

4.3. Selected candidates change often, selected answers much less often

Table 5 breaks the 4-bit and 3-bit changes down into answer transitions. With GPTQ-W4 or NF4 the selected candidate changes on 21–49% of main-study problems, but the selected canonical answer changes on only 4.8–13.2%,

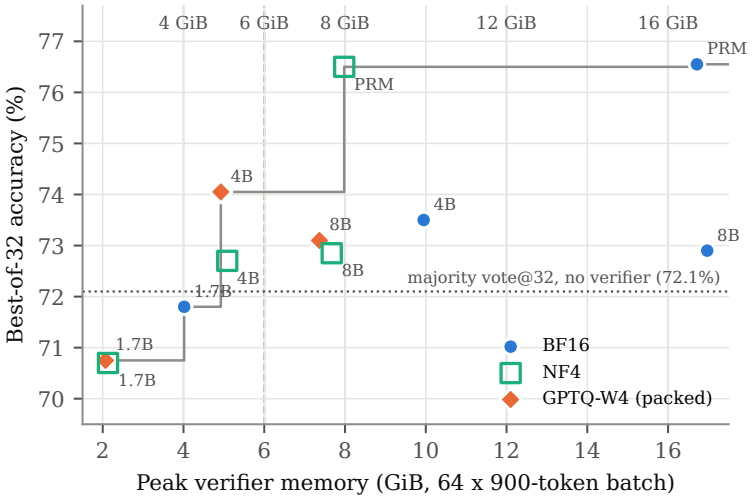


Figure 2: Accuracy against measured verifier memory for the tested Skywork ORM and PRM deployments (Best-of-32, main study, 2,000 problems; PRM memory from the exploratory re-measurement). Memory is the peak allocation while scoring a fixed workload of 64 sequences of 900 tokens on an RTX 4090 with no generator. GPTQ-W4 uses packed 4-bit kernels and NF4 uses `bitsandbytes`. Vertical lines mark the prespecified ceilings, and the dashed one is the 6 GiB ceiling of c2. The step line is the envelope of the tested neural-verifier deployments within each budget. Majority vote (dotted) needs no verifier memory and beats every tested verifier deployment below 4.9 GiB.

and 72–79% of the candidate changes keep the same final answer. Harm (a correct BF16 selection becomes wrong) and benefit (the reverse) each affect 0.9–4.1% of problems and largely cancel. At 3 bits the answer-change rate (14.2–38.9%) and harm both grow, and harm exceeds benefit. H6 and H7 show the same pattern. These measurements are consistent with answer redundancy in the pools. Candidates that share the selected answer are interchangeable for accuracy, so most score perturbations move the selection within the same answer cluster.

Table 5: How the Best-of-32 selection changes relative to BF16. Each row compares the quantized (or control) selection with the BF16 selection on the same problems. *cand.*: % of problems where a different candidate is selected; *answer*: % where the selected final answer changes; *same*: % of the candidate changes that keep the same final answer; *harm*: % where a correct BF16 selection becomes wrong; *benefit*: % where a wrong one becomes correct; Δ_{32} = benefit − harm (pp). When both winners lack an extractable answer we cannot compare their answers, so we count the switch as an answer change; the *answer* column is therefore an upper value, most of all for the ORMs that prefer prose answers.

study	verifier	condition	cand.	answer	same	harm	benefit	Δ_{32}
main	ORM 1.7B	GPTQ-W4	48.9	13.2	73	4.1	3.0	−1.10
		GPTQ-W3	87.6	38.9	56	17.7	4.5	−13.15
	ORM 4B	GPTQ-W4	29.2	8.1	72	1.7	2.3	+0.55
		GPTQ-W3	59.7	17.7	70	5.2	3.5	−1.75
	ORM 8B	BF16-null	6.9	1.0	86	0.3	0.2	−0.10
		NF4	21.2	4.8	78	1.3	1.2	−0.05
		GPTQ-W4	25.1	6.2	76	1.5	1.7	+0.15
		GPTQ-W3	50.5	14.2	72	4.3	2.7	−1.60
	PRM	NF4	25.2	5.7	78	0.9	0.9	−0.05
		GPTQ-W4	33.0	7.1	79	1.6	1.3	−0.30
		GPTQ-W3	54.0	14.3	74	3.4	2.1	−1.30
H6	ORM 8B	GPTQ-W4	31.4	6.9	78	1.3	1.3	0.00
	PRM	GPTQ-W4	34.4	5.8	83	0.8	1.3	+0.50
H7	ORM 8B	GPTQ-W4	26.9	8.8	67	2.1	1.8	−0.28
	PRM	GPTQ-W4	30.1	8.7	71	1.0	0.5	−0.57
	ORM 8B	GPTQ-W3	51.7	20.6	60	5.8	2.3	−3.49
	PRM	GPTQ-W3	55.9	20.4	64	3.8	1.2	−2.55

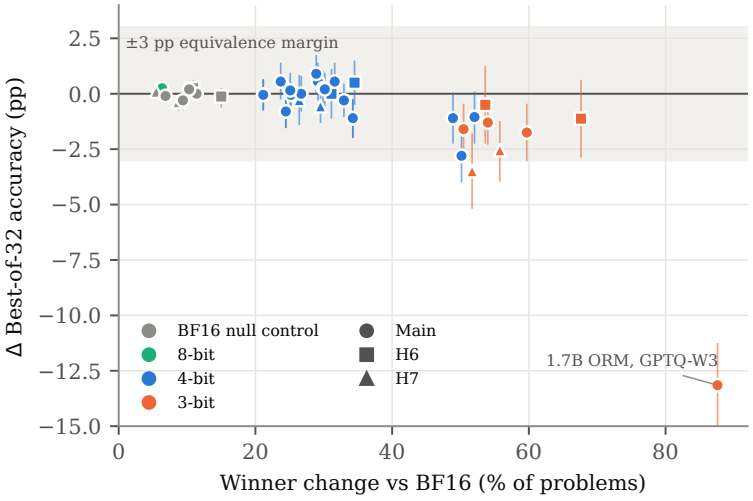


Figure 3: Winner change versus accuracy change at $N = 32$ for every quantized condition and control in the main study (circles), H6 (squares) and H7 (triangles), with 95% bootstrap intervals. The exploratory revision experiments (Llama-based ORMs, in-domain calibration) are not shown. Four-bit conditions change the selected candidate on 21–52% of problems, with point estimates inside the ±3 pp band. Three-bit conditions change about half or more of the winners and lose accuracy.

4.4. No confirmatory evidence for the MATH-loss and budget-growth contrasts

Development data had suggested that quantization damage concentrates on MATH and grows with N. On the 1,200 evaluation MATH problems, c4 (the MATH Δ_{32} of the 4B and 8B ORMs at GPTQ-W4, averaged within problem) is -0.63 pp $[-1.38, +0.08]$ with Holm $p = 0.40$. c5, the change of that loss from $N = 4$ to $N = 32$, is -0.51 pp $[-1.20, +0.16]$ with Holm $p = 0.42$. Neither hypothesis is supported. The results are inconclusive and do not show that the effect is zero, although the intervals rule out MATH losses larger than about 1.4 pp. c4 measures the loss within MATH and does not test an interaction with difficulty. The development signal came from 200

problems and did not replicate.

4.5. A stronger generator on hard problems (H6)

H6 tested whether the result holds with a stronger generator on harder, previously unused problems. Qwen2.5-Math-7B-Instruct [49] generated 32 candidates of up to 2,048 tokens for 800 MATH Level-5 test problems, which the 8B ORM and the PRM scored in BF16 and GPTQ-W4. The family of four tests asked for equivalence at ± 3 pp (H6a) and for retention of at least half of the BF16 verifier's advantage over majority vote (H6b). Let $G = A_{\mathrm{BF16}} - A_{\mathrm{MV}}$ be the BF16 advantage and Δ the quantization effect. H6b then tests

$$R \;=\; \tfrac{1}{2}G + \Delta \;=\; (A_{\mathrm{Q}} - A_{\mathrm{MV}}) - \tfrac{1}{2}G \;>\; 0,$$

that is, whether the quantized verifier's advantage exceeds half of the BF16 advantage. R is computed per problem as $R_i = C_{\mathrm{Q},i} - \frac{1}{2}(C_{\mathrm{BF16},i} + C_{\mathrm{MV},i})$ and bootstrapped as a mean. $R > 0$ implies retained value only when $G > 0$, so the plans interpret H6b and H7b only for verifiers with a positive BF16 advantage, which we check descriptively with 97.5% intervals.

Equivalence is supported for both verifiers, with $\Delta_{32} = 0.00$ pp $[-1.13, +1.13]$ for the ORM and $+0.50$ pp $[-0.50, +1.50]$ for the PRM, although the selected candidate changes on 31% and 34% of problems. With this generator, however, neither verifier showed a clear advantage over majority vote (76.5%). The PRM tied it ($G = +0.13$ pp $[-1.88, +2.13]$) and the ORM fell below it ($G = -3.75$ pp $[-6.25, -1.25]$). H6 therefore did not establish a positive BF16 advantage to retain, which limits what its retention tests can show (ORM $R = -1.88$ pp, PRM $+0.56$ pp; neither supported).

These conclusions hold under the lenient grader and without truncated candidates. Truncation is more common here than in the main study: 19% of candidates reached the length limit, all 32 candidates were truncated on 7 of the 800 problems, and at least three quarters were truncated on 58.

4.6. Value retention where the verifier adds value (H7)

H7 is a prospective study targeted at a regime chosen from earlier exploratory data, namely the main generator on hard problems, where the main-study PRM and 8B ORM had beaten majority vote. It uses all 1,060 MATH Level-4 and Level-5 test problems not used anywhere else in this work, the prespecified generation settings, and the same verifiers, conditions, tests and analysis code as H6. Because the regime was chosen with knowledge of earlier results, H7 estimates the effect of quantization in that regime. It does not estimate the typical value of a verifier across deployments.

On the new problems both verifiers beat majority vote in BF16, the PRM by $+8.2\,\text{pp}$ (97.5% interval $[+5.9, +10.7]$) and the 8B ORM by $+3.0\,\text{pp}$ $[+0.4, +5.8]$. All four prespecified hypotheses are supported (Holm over four). Accuracy is equivalent at $\pm 3\,\text{pp}$, with $\Delta_{32} = -0.28\,\text{pp}$ $[-1.42, +0.85]$ for the ORM and $-0.57\,\text{pp}$ $[-1.32, +0.19]$ for the PRM. Both quantized verifiers keep more than half of their BF16 advantage. The evidence is strong for the PRM ($R = +3.54\,\text{pp}$ $[+2.45, +4.72]$, Holm $p \leq 0.0003$) and narrow for the ORM ($R = +1.23\,\text{pp}$ $[-0.14, +2.59]$, one-sided Holm $p = 0.041$), in line with the ORM's lower planned power. Descriptively, the PRM retains 93% of its advantage (95% interval 84–102%) and the ORM 91% (44–149%). The ORM's interval is wide because its BF16 advantage, the denominator of the ratio, is small. The retention conclusions keep their sign under the lenient

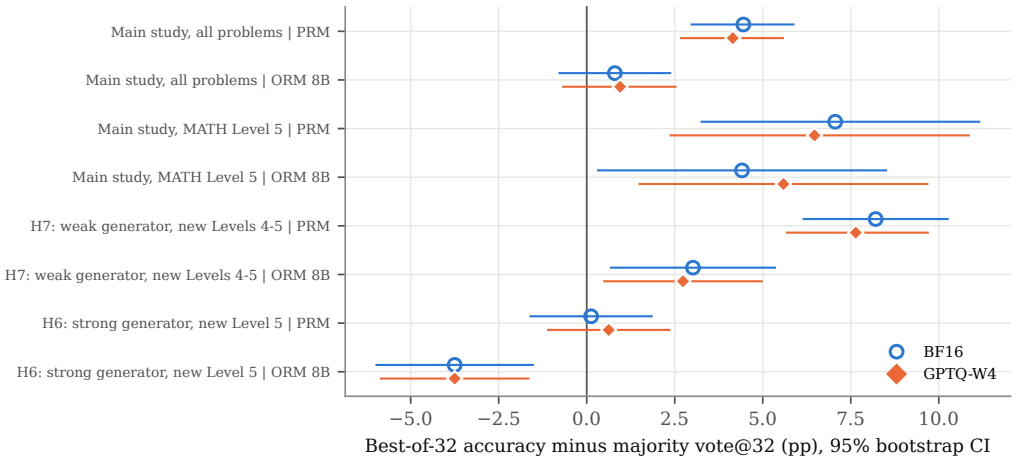


Figure 4: Verifier advantage over majority vote (Best-of-32 accuracy minus majority vote@32, pp, with 95% bootstrap intervals) in BF16 (open circles) and with GPTQ-W4 (diamonds), for the 8B ORM and the PRM. Main-study rows are exploratory subsets; H6 and H7 are separately planned studies on new problems.

grader and without truncated candidates, which the plan required before describing value as preserved. In an exploratory comparison, three bits is not safe in this regime: GPTQ-W3 costs 3.5 pp for the ORM and 2.6 pp for the PRM and removes the ORM's advantage. Level-wise estimates and all sensitivity analyses are in the repository, and Fig. 4 summarises the advantage over majority vote before and after quantization across regimes.

4.7. Robustness and controls

Other reward-model families. As an exploratory extension, we scored the previously analysed main and H7 pools with two further ORMs on a Llama 3.1 backbone (Table 6). The 4-bit result holds within the ±3 pp margin, with somewhat larger losses than for the Skywork-Qwen models. GPTQ-W4 changes the accuracy of Llama-3.1-Tulu-3-8B-RM by −0.60 pp on the main

problems and −0.75 pp on H7, and that of Skywork-Reward-V2-Llama-3.1-8B by −1.45 pp and −1.23 pp. All four 90% intervals lie inside ±3 pp, but the Skywork-Llama loss on the main problems is distinguishable from zero. Both models changed the selected candidate (44–53%) and the selected answer (11–25%) more often than the Skywork-Qwen ORMs. Neither showed a clear advantage G over majority vote in BF16, so value retention cannot be interpreted for them. With generic calibration at 3 bits, both Llama-based models break down. GPTQ-W3 costs Tulu-3 12.6 pp and Skywork-Llama 26.4 pp, which puts the latter below the random-candidate rate, against 1.6 pp for the Qwen-based 8B ORM. With generic calibration the 3-bit behaviour is therefore strongly model-dependent; for Tulu-3, in-domain calibration recovers most of the loss (next paragraph).

Table 6: Exploratory results for the two Llama-3.1-based ORMs (Best-of-32). A: BF16 accuracy (%); Δ: change with GPTQ-W4 (pp) with 95% and 90% intervals; G: BF16 advantage over majority vote (pp, 95% interval; majority vote 72.1% on the main problems and 55.9% on H7); cand./answer: share of problems whose selected candidate or answer changes (%). GPTQ-W3 with generic calibration on the main problems: Tulu-3 −12.60 $[-14.40, -10.75]$, answer change 35%; Skywork-Llama −26.35 $[-28.70, -24.05]$, answer change 59%.

ORM	population	A	Δ [95% CI]	90% CI	G [95% CI]	cand.	answer
Tulu-3-8B-RM	main (2,000)	72.5	−0.60 [−1.65, +0.45]	[−1.45, +0.30]	+0.35 [−1.20, +1.85]	44.6	11.5
	H7 (1,060)	55.0	−0.75 [−2.45, +0.85]	[−2.17, +0.57]	−0.94 [−3.11, +1.23]	43.6	18.3
Skywork-Llama-8B	main (2,000)	68.8	−1.45 [−2.75, −0.15]	[−2.55, −0.35]	−3.35 [−5.05, −1.65]	51.7	18.3
	H7 (1,060)	53.8	−1.23 [−3.11, +0.66]	[−2.74, +0.38]	−2.17 [−4.62, +0.28]	53.2	24.7

Task-matched calibration at 3 bits. Much of the 3-bit loss depends on the calibration data (exploratory; Table 7). The prespecified recipe uses 128

WikiText chunks of exactly 512 tokens. The in-domain alternative uses 128 complete candidate responses sampled from the generator's development pools, in the verifier's input format and of variable length (about 490–540 tokens on average, depending on the tokenizer). The calibration problems are development problems, disjoint from the evaluation set. With the in-domain recipe, the loss of the 1.7B ORM falls from 13.15 to 3.75 pp, and that of Tulu-3, a Llama-based ORM, from 12.60 to 2.80 pp. The recoveries are +9.40 pp and +9.80 pp. For the verifiers that lost little with generic calibration the change is smaller: +1.55 pp for the 8B ORM and +0.70 pp $[-0.35, +1.75]$ for the PRM, whose interval includes zero. A length control rules out sequence length as the explanation. Calibrating the 1.7B ORM on WikiText cut to exactly the token lengths of its in-domain set gives $\Delta_{32} = -16.75$ pp, no better than the generic recipe and 13.00 pp $[11.00, 14.95]$ worse than in-domain calibration. The recovery therefore comes from the content of the calibration text, its domain or its format, which this design does not separate. For the tested verifiers, the 3-bit boundary we observe depends on the calibration recipe as well as on the bit width. Skywork-Llama was not re-quantized with the in-domain recipe.

Selection rules. The 4-bit result does not depend on the selection rule (exploratory, main study, $N = 32$). In the reward-weighted answer vote each candidate with an extractable answer adds $\mathrm{softmax}(s/T)$ to its answer cluster, where s is the verifier score and T is fitted on development data in BF16 and kept fixed across precisions ($T = 0.3$ for the 8B ORM and 0.1 for the other verifiers). Under reward-weighted voting, BF16 accuracy and its change with GPTQ-W4 are 73.95% (−1.00 pp) for the 1.7B ORM, 76.00% (−0.60 pp) for

Table 7: GPTQ-W3 with generic and in-domain calibration (exploratory, main study, 2,000 problems, Best-of-32). Δ_{32} versus BF16 and the paired difference, in pp with 95% intervals. Length control for the 1.7B ORM: WikiText cut to the token lengths of its in-domain set, $\Delta_{32} = -16.75$ $[-18.75, -14.80]$.

verifier	BF16 A_{32} (%)	generic Δ_{32}	in-domain Δ_{32}	in-domain − generic
Skywork-1.7B	71.8	−13.15 [−15.05, −11.25]	−3.75 [−5.25, −2.30]	+9.40 [+7.55, +11.30]
Skywork-8B	72.9	−1.60 [−2.75, −0.45]	−0.05 [−1.00, +0.90]	+1.55 [+0.40, +2.75]
PRM-7B	76.6	−1.30 [−2.30, −0.25]	−0.60 [−1.45, +0.25]	+0.70 [−0.35, +1.75]
Tulu-3-8B-RM	72.5	−12.60 [−14.40, −10.75]	−2.80 [−4.00, −1.65]	+9.80 [+7.95, +11.60]

the 4B ORM, 76.20% (+0.30 pp) for the 8B ORM and 79.50% (+0.25 pp) for the PRM. Restricting Best-of-32 to candidates with an extractable answer gives 73.95% (−1.10 pp), 75.95% (−0.55 pp), 76.10% (0.00 pp) and 79.75% (−0.45 pp) for the same four verifiers, and last-step aggregation for the PRM gives 79.35% (−0.25 pp). GPTQ-W3 loses 0.7–11 pp under every rule. Restricting selection to extractable answers raises the BF16 ORMs' accuracy by 2.2–3.2 pp (for the 8B ORM, from 72.9% to 76.1%), which shows how much the strict grader penalises their preference for prose answers. Under both rules the c2 contrast stays positive (+1.40 and +1.50 pp; Section 4.2).

Answer format. Under the strict grader several quantized conditions appeared to *improve* GSM8K accuracy, for example by +3.0 pp for the 4B ORM at GPTQ-W4 and by +8.0 pp on the Llama pools. The format/content decomposition (Section 3.6) shows that these gains come from answer format. Quantization weakens the Skywork ORMs' preference for prose answers without a boxed final answer. For the 4B ORM at GPTQ-W4 the format term is +3.0 pp and the content term 0.0 pp, and the difference under the lenient grader is +0.4 pp. On MATH the changes come from answer content

under both graders. We therefore claim no accuracy gain from quantization.

Packed versus simulated quantization. Packed GPTQ-W4, which runs FP16 Marlin kernels, and simulated GPTQ-W4, which runs in BF16, select the same final answer on 98.8–99.2% of problems and the same candidate on 84.7–88.8%. Their candidate-level disagreement (11.3–15.4%) is somewhat larger than the bf16-null floor (6.9–11%), and their answer-level disagreement (0.8–1.2%) is of the same size as the null control's (1.0% for the 8B ORM). Correctness flips are rare and symmetric (0.35–0.55% of problems). Packed GPTQ-W4 changes accuracy versus BF16 by -1.05, $+0.55$ and $+0.20$ pp for the 1.7B, 4B and 8B ORMs, against -1.10, $+0.55$ and $+0.15$ pp for simulated quantization. c2 uses the packed scores.

Second generator. On 300 problems with Llama-3.2-3B-Instruct candidates (exploratory), GPTQ-W4 changes the PRM's Best-of-32 accuracy by $+0.33$ pp $[-3.00, +4.00]$ and NF4 by -0.67 pp. For the 4B ORM the strict grader shows $+8.0$ pp, which shrinks to $+2.0$ pp $[0.0, +4.3]$ under the lenient grader, again a format effect. With this generator, majority vote (76.3%) beats both verifiers in BF16 (ORM 63.7%, PRM 70.7%).

Labels and truncation. With GSM8K-Platinum labels the c1 estimate is $+0.44$ pp. c4 and c5 involve MATH only and do not change. With truncated candidates removed before selection, the estimates are $+0.33$ pp for c1, -0.67 pp for c4 and -0.52 pp for c5. None of these sensitivity estimates reverses the direction of a prespecified conclusion. Both GSM8K problems whose labels the human audit had flagged as questionable are among those that GSM8K-Platinum removes.

5. Discussion

Deployment implications. For the tested reward-model verifiers with 4B or more parameters, 4-bit weight-only GPTQ preserved Best-of-N accuracy in three prospectively planned settings, which covered two generators, three problem sets, an ORM family and a PRM. Where the verifier beat majority vote, the quantized verifier kept most of that advantage. NF4 and AWQ behaved similarly in the exploratory grid but were not part of H6 and H7. Two reward models on a Llama backbone also stayed within the margin. They lost 0.6–1.5 pp and changed their selections more often, though, so the size of the 4-bit effect varies between models even when it is small. Under the tested 6 GiB verifier budget and workload, a 4B ORM with packed 4-bit weights was the best tested deployment, ahead of the best full-precision verifier that fit.

Three-bit GPTQ with generic calibration was not robust. It cost 1.3–1.8 pp for the larger verifiers in the main study, 2.6–3.5 pp in H7, 12.6–26.4 pp for the two Llama-based reward models and 13.2 pp for the 1.7B ORM, which was also sensitive to the choice between RTN and GPTQ at 4 bits. In exploratory tests, calibrating on in-domain candidates instead of generic text substantially reduced this loss for both models that had failed, from -13.15 to -3.75 pp for the 1.7B ORM and from -12.60 to -2.80 pp for Tulu-3. The 8B ORM and the PRM, which lost little to begin with, moved to -0.05 and -0.60 pp. A length-matched generic control did not help, so the content of the calibration text matters, although we cannot say whether its domain or its format is responsible. At 3 bits, the calibration recipe matters as much as the bit width. All of these conclusions are specific to the tested checkpoints, quantization

recipes, scoring rule and mathematical tasks.

Why accuracy is stable when selections change. At 4 bits quantization perturbs verifier scores enough to change the selected candidate on a quarter to a half of problems, yet the selected canonical answer changes far less often (Table 5). Most changes stay within the same answer cluster, and the remaining harms and benefits largely cancel. This is consistent with answer redundancy in the pools. For Best-of-N to stay correct, the top of the ranking has to keep a correct answer, and any of the candidates that carry it will do. We have not isolated this mechanism causally, for example by manipulating answer multiplicities, and we have not measured where the remaining answer-level changes occur. A testable implication is that quantization sensitivity depends on how closely answers compete within a pool more than on nominal difficulty or on N.

How much a verifier is worth. Testing whether quantization preserves a verifier's advantage over majority vote requires a setting where the full-precision verifier has one. With the main generator, the BF16 Skywork ORMs were within -0.3 to $+1.4$ pp of majority vote. With the strong math generator on hard problems (H6), majority vote matched the PRM and beat the 8B ORM. Both verifiers clearly beat majority vote only in H7, the regime we had selected for verifier advantage. Aggregation rules that combine answer frequency and reward scores can change this comparison [19], and our weighted-vote control shows the same robustness to quantization under such a rule. Practitioners should compare a verifier with self-consistency on their own generator before paying its memory and latency cost.

Evaluating selection. Three practices changed our conclusions or the size of our estimates. First, a strict answer grader turned a reward-model preference for prose answers into apparent quantization "gains" of up to 8 pp. A format/content decomposition and a lenient grader substantially reduced these gains, and a blinded human audit of the winners without an extractable answer reduced the prespecified memory-budget effect from 2.25 to 1.4 pp. Second, the selected candidate changes even without quantization. Rescoring the same BF16 weights with a different batch layout changes 5–15% of selected candidates across our studies, because small numerical differences are enough to reorder near-tied candidates. A winner-change rate is therefore only informative next to such a baseline: without it, quantization would look more disruptive than it is. Third, two effects seen on development problems (larger damage on MATH and damage growing with N) did not replicate on 1,200 evaluation problems. Because we had stated them as hypotheses before the evaluation, this non-replication is reported as such instead of being lost among exploratory results.

Limitations. The main-study ORMs come from a single family (Skywork-Reward-V2 on Qwen3), the PRM is also Qwen-based, and the generators are Qwen and Llama models. The two additional Llama-based reward models vary the backbone and the training family only in exploratory runs. Neither beat majority vote in BF16, so value retention is untested outside the Qwen-based verifiers. Accuracy experiments use simulated quantization. Packed 4-bit kernels agreed with it on 98.8–99.2% of selected answers, and memory and throughput were measured for one scoring workload without the generator. The confirmatory GPTQ results use generic calibration with a

single calibration draw, and in-domain calibration was tested only at 3 bits, for four verifiers and with one calibration draw. The tasks are GSM8K and MATH, whose test problems may have been seen during pre-training of the generators and verifiers. Because BF16 and quantized verifiers score the same frozen pools, any exposure is shared by design, but it may alter pool redundancy and verifier advantage and could interact with quantization. Candidates were capped at 1,024 or 2,048 tokens and 6.5–19% reached the cap, although excluding them changes no conclusion. The human audit covered all winners without an extractable answer plus 30 controls with one, and it was done by one rater, so inter-rater agreement could not be measured and the audit-corrected estimate does not include rater uncertainty. The value-retention result is confirmatory for the PRM and only narrowly so for the ORM, and it holds in a regime chosen with prior knowledge. We did not find a strong-generator regime in which the verifiers clearly beat majority vote. Finally, the analysis plans were fixed in version control without an external registry, and the main-study plan was amended after candidate generation but before any verifier scoring.

6. Conclusion

We evaluated whether reward-model verifiers in Best-of-N mathematical answer selection can be quantized without losing selection accuracy or their advantage over majority vote. On frozen candidate pools and with analysis plans fixed in advance, the tested 4-bit GPTQ deployments of verifiers with 4B or more parameters kept Best-of-32 accuracy within a ± 3 pp equivalence margin in the main study and in two further studies (H6 and H7). Where the

verifier beat majority vote, the quantized verifier kept more than half of that advantage, an estimated 93% for the PRM. Under the tested 6 GiB verifier budget, the 4B ORM with packed 4-bit weights was the best tested deployment. Quantization often changes the selected candidate but changes the selected answer much less often, which is consistent with answer redundancy in the pools. Three-bit GPTQ with generic calibration degrades selection, most for the smallest Qwen-based ORM and severely for both Llama-based reward models. Calibrating on in-domain candidates recovers most of the loss for both verifiers that failed among those we re-quantized (the 1.7B ORM and Tulu-3), and a length-matched generic control does not.

These results carry two practical lessons. First, a verifier should be compared with majority vote on the target generator, because with a strong generator it may add little. Second, selection should be evaluated with a grader that is robust to answer format, because a strict grader can turn format preferences into spurious effects of quantization.

CRediT authorship contribution statement

Hector Borobia: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elies Seguí-Mas:** Writing – review & editing, Supervision. **Guillermina Tormo-Carbó:** Writing – review & editing, Supervision.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code, configuration, frozen problem manifests, analysis plans with their amendments, analysis scripts and figure-generation scripts are available at `https://github.com/hecboar/quantized-verifiers-bon` (tag `v1.0-submission`). The candidate pools, all verifier scores and the result tables are available at `https://huggingface.co/datasets/hecboar/quantized-verifiers-bon`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Code (Anthropic) to support code development, language editing, and consistency checking. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, J. Schulman, Training verifiers to solve math word problems, arXiv preprint arXiv:2110.14168 (2021).

[2] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, K. Cobbe, Let's verify step by step, in: The Twelfth International Conference on Learning Representations (ICLR), 2024, arXiv:2305.20050.

[3] C. Snell, J. Lee, K. Xu, A. Kumar, Scaling LLM test-time compute optimally can be more effective than scaling parameters for reasoning, in: The Thirteenth International Conference on Learning Representations (ICLR), 2025, preprint title: "Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters", arXiv:2408.03314.

[4] E. Frantar, S. Ashkboos, T. Hoefler, D. Alistarh, GPTQ: Accurate post-training quantization for generative pre-trained transformers, in: The Eleventh International Conference on Learning Representations (ICLR), 2023, published at ICLR 2023 as "OPTQ: Accurate Quantization for Generative Pre-trained Transformers"; arXiv:2210.17323.

[5] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, S. Han, AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration, in: Proceedings of Machine Learning and Systems, Vol. 6, 2024, arXiv:2306.00978.

[6] T. Dettmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer, QLoRA: Efficient finetuning of quantized LLMs, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 10088–10115.

[7] L. Gao, J. Schulman, J. Hilton, Scaling laws for reward model overoptimization, in: Proceedings of the 40th International Conference on Machine Learning (ICML), Vol. 202 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 10835–10866.

[8] P. Wang, L. Li, Z. Shao, R. X. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, Z. Sui, Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations (2023). arXiv:2312.08935.

[9] A. Setlur, C. Nagpal, A. Fisch, X. Geng, J. Eisenstein, R. Agarwal, A. Agarwal, J. Berant, A. Kumar, Rewarding Progress: Scaling Automated Process Verifiers for LLM Reasoning (2024). arXiv:2410.08146.

[10] Z. Zhang, C. Zheng, Y. Wu, B. Zhang, R. Lin, B. Yu, D. Liu, J. Zhou, J. Lin, The lessons of developing process reward models in mathematical reasoning, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria, 2025, pp. 10495–10516, arXiv:2501.07301. doi:10.18653/v1/2025.findings-acl.547.

[11] L. Zhang, A. Hosseini, H. Bansal, M. Kazemi, A. Kumar, R. Agarwal, Generative Verifiers: Reward Modeling as Next-Token Prediction (2024). arXiv:2408.15240.

[12] R. Liu, J. Gao, J. Zhao, K. Zhang, X. Li, B. Qi, W. Ouyang, B. Zhou, Can 1B LLM Surpass 405B LLM? Rethinking Compute-Optimal Test-Time Scaling (2025). arXiv:2502.06703.

[13] N. Singhi, H. Bansal, A. Hosseini, A. Grover, K.-W. Chang, M. Rohrbach, A. Rohrbach, When To Solve, When To Verify: Compute-Optimal Problem Solving and Generative Verification for LLM Reasoning (2025). arXiv:2504.01005.

[14] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: The Eleventh International Conference on Learning Representations (ICLR), 2023, arXiv:2203.11171.

[15] B. Brown, J. Juravsky, R. Ehrlich, R. Clark, Q. V. Le, C. Ré, A. Mirhoseini, Large Language Monkeys: Scaling Inference Compute with Repeated Sampling (2024). arXiv:2407.21787.

[16] Y. Wu, Z. Sun, S. Li, S. Welleck, Y. Yang, Inference Scaling Laws: An Empirical Analysis of Compute-Optimal Inference for Problem-Solving with Language Models (2024). arXiv:2408.00724.

[17] L. Yuan, W. Li, H. Chen, G. Cui, N. Ding, K. Zhang, B. Zhou, Z. Liu, H. Peng, Free Process Rewards without Process Labels (2024). arXiv:2412.01981.

[18] Q. Di, K. Ji, X. Li, H. Zhao, Q. Gu, Best-of-Majority: Minimax-Optimal Strategy for Pass@k Inference Scaling (2025). arXiv:2510.03199.

[19] P. Kuang, Y. Wang, X. Han, Y. Liu, K. Xu, H. Wang, Optimal aggregation of LLM and PRM signals for efficient test-time scaling, arXiv preprint arXiv:2510.13918 (2025).

[20] J. Kim, E. Ewer, T. Moon, J. Park, D. Papailiopoulos, Not All Bits Are Equal: Scale-Dependent Memory Optimization Strategies for Reasoning Models, in: The Fourteenth International Conference on Learning Representations (ICLR), 2026, arXiv:2510.10964.

[21] H. Sun, M. Haider, R. Zhang, H. Yang, J. Qiu, M. Yin, M. Wang, P. Bartlett, A. Zanette, Fast best-of-N decoding via speculative rejection, in: Advances in Neural Information Processing Systems, 2024, arXiv:2410.20290.

[22] J. Guo, Z. Wu, H. Yang, P. S. Yu, Mining intrinsic rewards from LLM hidden states for efficient best-of-N sampling, arXiv preprint arXiv:2505.12225 (2025).

[23] P. Kuang, H. Jin, X. Han, Y. Wang, X. Yuan, Y. Yu, K. Xu, H. Wang, KV-PRM: Efficient process reward modeling via KV-cache transfer for multi-agent test-time scaling, arXiv preprint arXiv:2607.09153 (2026).

[24] N. Lambert, V. Pyatkin, J. Morrison, L. Miranda, B. Y. Lin, K. Chandu, N. Dziri, S. Kumar, T. Zick, Y. Choi, N. A. Smith, H. Hajishirzi, RewardBench: Evaluating Reward Models for Language Modeling (2024). arXiv:2403.13787.

[25] C. Zheng, Z. Zhang, B. Zhang, R. Lin, K. Lu, B. Yu, D. Liu, J. Zhou,

J. Lin, ProcessBench: Identifying Process Errors in Mathematical Reasoning (2024). arXiv:2412.06559.

[26] M. Song, Z. Su, X. Qu, J. Zhou, Y. Cheng, PRMBench: A Fine-grained and Challenging Benchmark for Process-Level Reward Models (2025). arXiv:2501.03124.

[27] T. Dettmers, M. Lewis, Y. Belkada, L. Zettlemoyer, LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale (2022). arXiv:2208.07339.

[28] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, S. Han, SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models (2022). arXiv:2211.10438.

[29] T. Dettmers, L. Zettlemoyer, The case for 4-bit precision: k-bit Inference Scaling Laws (2022). arXiv:2212.09720.

[30] E. Kurtic, A. Marques, S. Pandit, M. Kurtz, D. Alistarh, "Give Me BF16 or Give Me Death"? Accuracy-Performance Trade-Offs in LLM Quantization (2024). arXiv:2411.02355.

[31] C. Zhu, H. Jiang, J. Liu, Y. Han, K. Zhao, R. Zhang, RotSQ: Joint quantization and sparsification of large language models in rotated space, Neurocomputing 707 (2027) 135039. doi:10.1016/j.neucom.2026.135039.

[32] W. Bang, J. Park, H. Yu, S. Han, K. Bin, K. Lee, BVSR: Bitvector-sum representation for distribution-aware LLM weight quantization, Neurocomputing 709 (2027) 135315. doi:10.1016/j.neucom.2026.135315.

[33] Z. Yang, S. Hou, K. Zheng, Y. Liu, L. He, Y. Chen, K. Cui, Target-aware calibration data selection for preserving uncertainty in quantized language models (2026). arXiv:2608.21019.

[34] R. Liu, Y. Sun, M. Zhang, H. Bai, X. Yu, T. Yu, C. Yuan, L. Hou, Quantization Hurts Reasoning? An Empirical Study on Quantized Reasoning Models (2025). arXiv:2504.04823.

[35] Z. Li, Y. Su, R. Yang, C. Xie, Z. Wang, Z. Xie, N. Wong, H. Yang, Quantization Meets Reasoning: Exploring LLM Low-Bit Quantization Degradation for Mathematical Reasoning (2025). arXiv:2501.03035.

[36] A. Dutta, S. Krishnan, N. Kwatra, R. Ramjee, Accuracy is Not All You Need (2024). arXiv:2407.09141.

[37] M. Patira, When BitsAndBytes changes the evaluator: Reliability of 8/4-bit LLM judges, in: NeurIPS 2026 Workshop LIGHT, 2026, oral. OpenReview https://openreview.net/forum?id=3ChXlWOIrY.

[38] H. Zhang, J. Da, D. Lee, V. Robinson, C. Wu, W. Song, T. Zhao, P. Raja, C. Zhuang, D. Slack, Q. Lyu, S. Hendryx, R. Kaplan, M. Lunati, S. Yue, A Careful Examination of Large Language Model Performance on Grade School Arithmetic (2024). arXiv:2405.00332.

[39] J. Vendrow, E. Vendrow, S. Beery, A. Madry, Do large language model benchmarks test reliability?, arXiv preprint arXiv:2502.03461 (2025).

[40] A. Hochlehnert, H. Bhatnagar, V. Udandarao, S. Albanie, A. Prabhu, M. Bethge, A Sober Look at Progress in Language Model Reasoning: Pitfalls and Paths to Reproducibility (2025). arXiv:2504.07086.

[41] S. Biderman, H. Schoelkopf, L. Sutawika, L. Gao, J. Tow, B. Abbasi, A. F. Aji, P. S. Ammanamanchi, S. Black, J. Clive, A. DiPofi, J. Etxaniz, B. Fattori, J. Z. Forde, C. Foster, J. Hsu, M. Jaiswal, W. Y. Lee, H. Li, C. Lovering, N. Muennighoff, E. Pavlick, J. Phang, A. Skowron, S. Tan, X. Tang, K. A. Wang, G. I. Winata, F. Yvon, A. Zou, Lessons from the Trenches on Reproducible Evaluation of Language Models (2024). arXiv:2405.14782.

[42] E. Miller, Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations (2024). arXiv:2411.00640.

[43] J. M. Hofman, A. Chatzimparmpas, A. Sharma, D. J. Watts, J. Hullman, Pre-registration for Predictive Modeling (2023). arXiv:2311.18807.

[44] D. Hendrycks, C. Burns, S. Kadavath, A. Arora, S. Basart, E. Tang, D. Song, J. Steinhardt, Measuring mathematical problem solving with the MATH dataset, in: Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks, Vol. 1, 2021, arXiv:2103.03874.

[45] Qwen Team, A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al., Qwen2.5 technical report, arXiv preprint arXiv:2412.15115 (2024).

[46] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with PagedAttention, in: Proceedings of the 29th Symposium on Operating Systems Principles (SOSP), ACM, 2023, pp. 611–626. doi:10.1145/3600006.3613165.

[47] C. Y. Liu, L. Zeng, Y. Xiao, J. He, J. Liu, C. Wang, R. Yan, W. Shen, F. Zhang, J. Xu, Y. Liu, Y. Zhou, Skywork-Reward-V2: Scaling preference data curation via human-AI synergy, in: The Fourteenth International Conference on Learning Representations (ICLR), 2026, arXiv:2507.01352.

[48] N. Lambert, J. Morrison, V. Pyatkin, S. Huang, H. Ivison, F. Brahman, L. J. V. Miranda, A. Liu, N. Dziri, S. Lyu, et al., Tülu 3: Pushing frontiers in open language model post-training, arXiv preprint arXiv:2411.15124 (2024).

[49] A. Yang, B. Zhang, B. Hui, B. Gao, B. Yu, C. Li, D. Liu, J. Tu, J. Zhou, J. Lin, et al., Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement, arXiv preprint arXiv:2409.12122 (2024).

Appendix A. Implementation details

Prompt. Every generator receives the system prompt "Please reason step by step, and put your final answer within `\boxed{}`." and the problem as the user turn, through its chat template. For Llama-3.2-3B the instruction is repeated at the end of the user turn. This is a recorded deviation: without the repetition, 34% of its candidates had no extractable answer.

Verifier inputs. The ORMs (Skywork-Reward-V2-Qwen3 1.7B/4B/8B, Skywork-Reward-V2-Llama-3.1-8B, Llama-3.1-Tulu-3-8B-RM) receive the (problem, response) pair as a user/assistant conversation formatted by their

chat template, without a system prompt and with a single beginning-of-sequence token. They output a scalar from their reward head, and we verified that the padding token never appears inside a sequence. Tulu-3-8B-RM does not define a padding token ID in its model configuration, so we assign the tokenizer's padding token ID. The PRM receives its documented system prompt, the problem, and the response split into steps at blank lines and separated by its step token. Its score is the minimum step reward, and the last-step reward is stored for the sensitivity analysis.

Quantization. All linear layers inside the transformer blocks are quantized (weight-only), while embeddings, norms, the reward head and activations stay in BF16. GPTQ uses group size 128, symmetric quantization and 1% diagonal damping, and it is calibrated on 128 WikiText-2 sequences of 512 tokens drawn with a fixed seed. AWQ uses the same calibration and group size, NF4 uses `bitsandbytes` without double quantization, and RTN uses group size 128 at 4 bits and per-channel scales at 8 bits. We audit every quantized model by fingerprinting all parameters before and after quantization. Every target layer must change (for example 252 of 252 for the 8B ORM and 196 of 196 for the PRM), and no other parameter may change.

PRM rotary buffers. With `transformers` 5.x, the remote modelling code of Qwen2.5-Math-PRM-7B is instantiated on the meta device, and its non-persistent rotary-embedding buffers come back uninitialised after loading. Scores computed this way were below the random-candidate rate. We recompute the buffers exactly as the model's own initialisation does and check them at every load. All PRM results in this paper use repaired buffers, and the affected scores were discarded (a recorded deviation).

Software and compute. Generation used vLLM 0.30.0. Scoring used PyTorch 2.13.0 (CUDA 13.0) with `transformers` 5.17 in the main study and 5.19 in H6 and H7, and the packed kernels used `gptqmodel`. Model identifiers and all settings are recorded in the repository and in the metadata of every score file. All experiments ran on single cloud GPUs, mostly RTX 4090, with some H6 and H7 runs on L40S and other GPUs with at least 24 GB. The BF16 and quantized runs of a verifier always used the same GPU. The total was roughly 90 GPU-hours, including setup.

Appendix B. Blinded audit of the outcome measure

A human rater (the first author) inspected all 94 distinct Best-of-32 winners without an extractable final answer in either arm of the memory-budget contrast c2 (48 from the 4B GPTQ-W4 arm and 68 from the 1.7B BF16 arm, with 22 shared), together with 30 randomly sampled winners that had an extractable answer. Each item showed only the problem, the reference answer and the full response, typeset as mathematics. The endpoint was final-answer correctness. Responses that reached the length limit were marked as such and checked individually, and by a rule fixed beforehand they counted as incorrect when they stopped before stating a final answer. Of the 24 truncated responses audited, 23 were judged incorrect. The remaining one (R079) had stated its final answer before the cut and was judged correct.

The rater judged 83 responses correct, 29 incorrect and 12 unsure. In a blinded second pass, with the same view and before opening the key, the rater re-examined every unsure and incorrect verdict. The pass confirmed all 29 incorrect verdicts and resolved all 12 unsure ones as incorrect: 11 had

been cut by the length limit before giving a final answer, 3 of them on a correct track, and one had a wrong final answer. It also flagged two GSM8K problems with questionable reference labels, both of which GSM8K-Platinum removes. If the first-pass verdicts are used and the unsure items dropped, the corrected c2 is $+1.41$ pp $[+0.35, +2.46]$ instead of $+1.40$. On the 30 controls the strict grader and the human agree on 29. On the 94 winners without an extractable answer the strict grader scores all as wrong, while the human judges 60 correct (27 of 48 in the 4B arm, 45 of 68 in the 1.7B arm and 12 of the 22 shared), and the lenient grader agrees with the human on 85. The audit covers all such winners plus 30 controls and was done by a single rater, so grading error among winners with an extractable answer is measured only on the 30 controls.

Appendix C. Analysis plans and deviations

Table C.8 lists the chronology recorded in the repository. All deviations are documented with dates. In the main study, the formatting instruction was repeated in the Llama prompt, and the PRM rotary buffers were repaired before rescoring. The prespecified decision rule permitted running c3, whose calibration hyper-parameter had been frozen on development data. At the prespecified 4 bits, however, the 8B ORM showed no MATH damage to repair, so c3 was not run and stays in the family with $p = 1$. After the main results we added the exploratory bit-width dose-response runs and included the packed 1.7B model in the memory frontier. In H6 the first candidate pool was lost before inspection and regenerated, and scoring was split across four GPUs. In H7, GPUs other than the RTX 4090 had to be used because of limited

availability, and the lenient-grader verdicts were computed with time limits equivalent to those on Linux. The corrections to the H6/H7 analysis script were made before H7 data existed and reproduce all H6 results exactly. The exploratory revision experiments (additional reward models, task-matched 3-bit calibration, and the memory and throughput re-measurement), and the second set (in-domain 3-bit calibration for the PRM and Tulu-3, and the length control), were specified before their launch.

One analysis correction was made after the main results. The exploratory majority-vote baseline of the main study had broken ties between answer clusters by verifier score, and it fell back to the Best-of-N winner when no candidate had an extractable answer. On 14 October it was replaced by the verifier-independent rule of Section 3.5. The change affects the majority outcome on 12–14 of the 2,000 main-study problems (72.4–72.7% becomes 72.1%) and only exploratory main-study tables and figures. The H6 and H7 analyses always used the verifier-independent rule, so no confirmatory test, gain G or retention statistic R changes.

Table C.8: Chronology of plans and data (commit timestamps, 2026).

date	event	data existing at that point
27 Sep	plan (c1–c3); manifests frozen	none
27–28 Sep	go/no-go gate on development problems	dev. scores
28 Sep 11:20	amendment 1: c4, c5, MATH extension	dev. scores
28 Sep 11:43–14:00	main-study pools generated	candidates[a]
28 Sep 14:17	amendment 2: populations, tests, c3 rule, hashes	candidates[a]
from 28 Sep 14:17	main-study scoring and analysis	–
12 Oct 19:00	H6 plan, manifests, analysis code	none
13 Oct 02:10	H6 pool hashes; then scoring	candidates
13 Oct 13:17	H7 plan (same analysis code)	none
13 Oct 14:27	H7 pool hashes; then scoring	candidates
14 Oct 00:08	majority-vote tie rule corrected (exploratory tables)	all main scores
14 Oct 00:14	revision experiments fixed (exploratory)	none of the new runs

[a] Generator statistics only (accuracy, extractable answers, truncation); no verifier score.